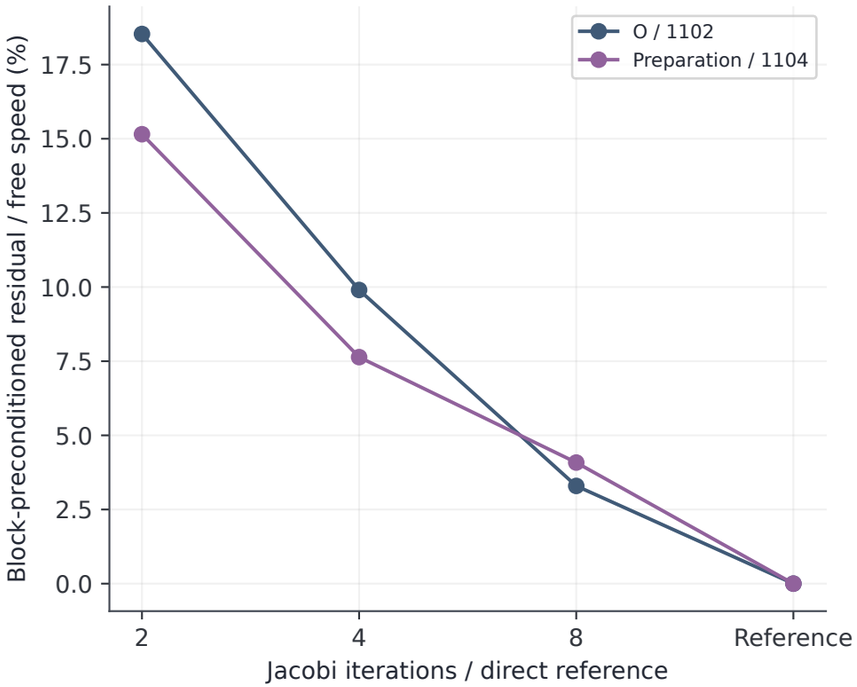
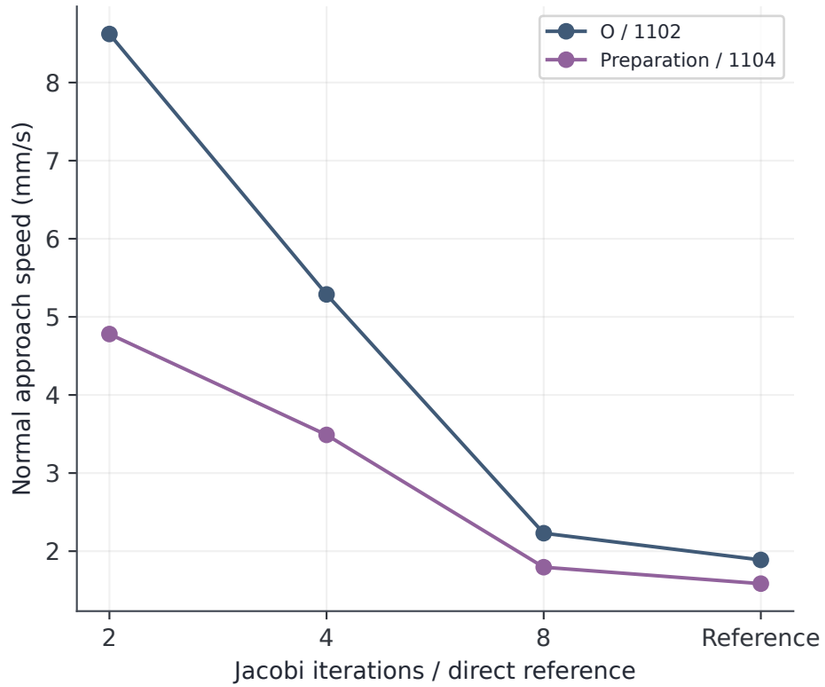
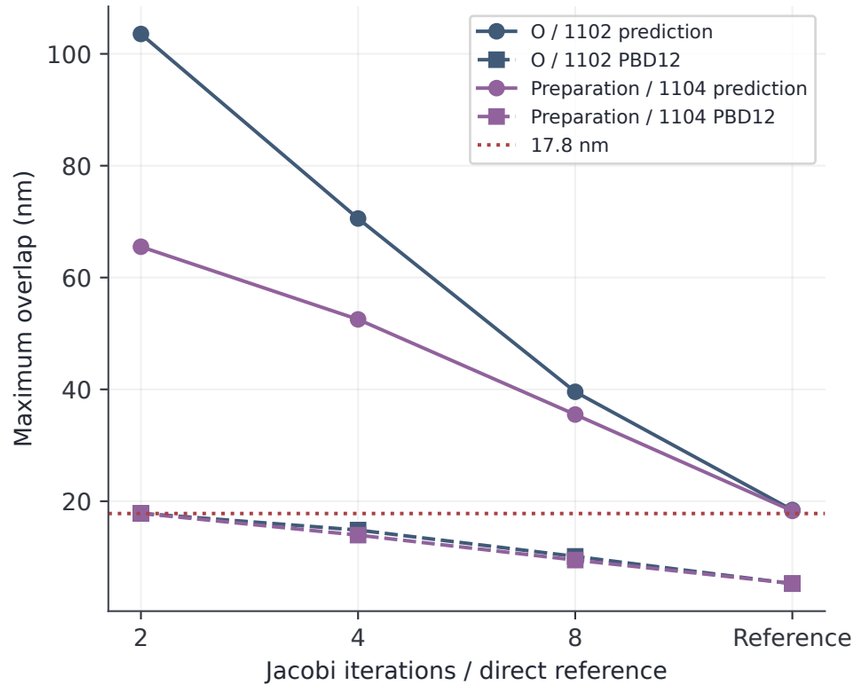
a Frozen-state algebraic residual (F10G reused)



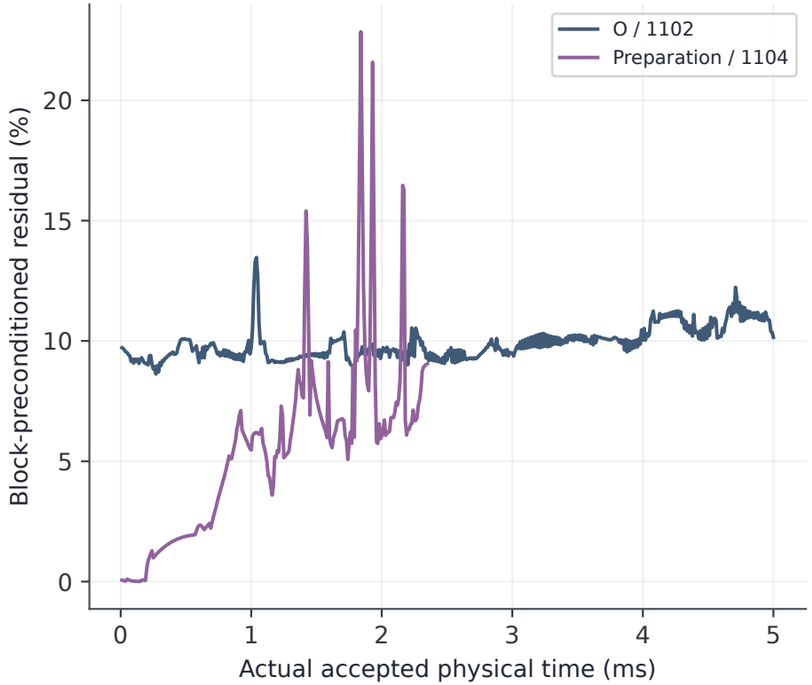
b Frozen-state maximum approach speed



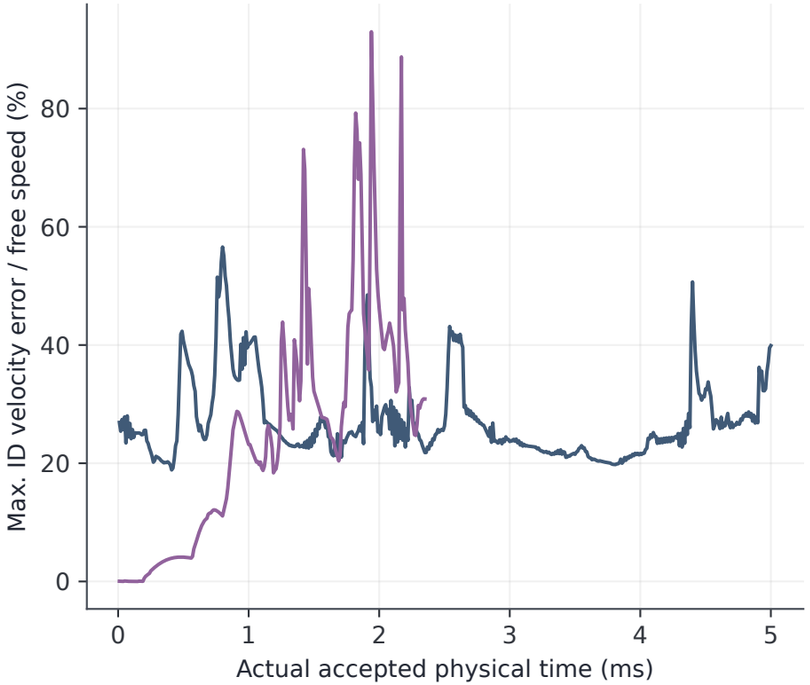
c Frozen-state overlap and original budget



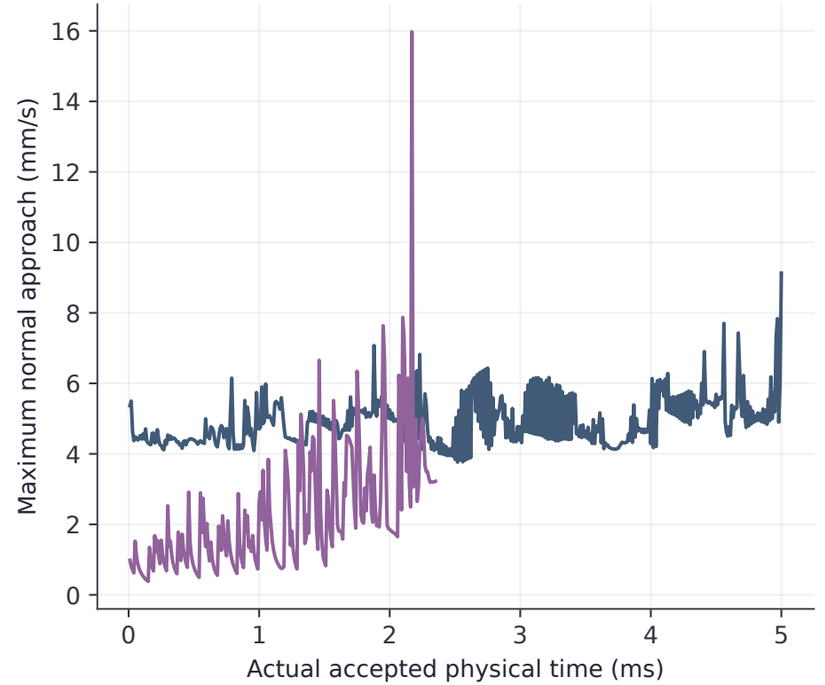
d Actual LUB4 trajectory residual



e Actual LUB4 discrete-reference difference



f Actual LUB4 normal approach speed



Top: identical saved positions, forces and 10 µs step; reused 2/4/8/direct resistance references. Bottom: genuinely evolved LUB4 trajectories.
Algebraic residuals refer to the existing approximate resistance model; they are not measured physical fluid-velocity errors.